Cluster Analysis Notes

Introduction

The objective of cluster analysis (CA) is to uncover structure in a dataset by aggregating similar rows into groups called clusters. Therefore, CA contrasts with PCA (principal components analysis)[1] in finding simplification among the rows instead of among the columns of the dataset. CA and PCA both attempt to simplify a dataset and do so without supervision. However, PCA actually can reduce the number of columns, whereas CA does not actually reduce the number of rows. Moreover, PCA changes the values of the variables, whereas CA does not change the values of any variable. But for some purposes, each cluster may be treated as if each were a single entity (as in a marketing campaign, for example) and thus CA effectively reduces the number of entities to understand. Grouping in CA is based on various measures of similarity or dissimilarity among the rows. Good clusters are those that have very similar rows within each cluster but very different rows between clusters.

The two major problems of CA are to determine the number of clusters and to assign the rows to clusters.

Some examples of applications of CA:

- Marketing: Market segmentation
- Politics: Analysis of voter blocs
- Psychology, psychiatry: Definition of personality types, mental diseases
- Biology: Reconstructing the evolutionary "tree of life"
- Linguistics: Reconstructing ancestral languages
- Anthropology: Mapping patterns of human migration and settlement
- Environment: Identifying weather patterns and their propensity for air pollution

You should understand clearly that there is *no supervision* in CA. What does that mean? It means that there is no training set of rows that have been correctly assigned to clusters, from which one can infer a rule for assigning unclassified rows to clusters. You do not know how many clusters there are, nor do you know *any* members of *any* cluster. Yet you must still figure out how many clusters there are and assign each row to a cluster.

There are parallel *supervised* statistical techniques that do infer rules to assign rows to clusters. If you know how many groups there are and you have a training set of rows that have been correctly assigned to groups, you have supervision. Then you can infer rules for future assignment of rows to groups based upon the characteristics of the training set rows that have been correctly assigned to each group. Supervised assignment of rows to groups is called *classification*. Two examples of classification methodologies are discriminant analysis and logistic regression.

> Example: Suppose that an insurance company would like to develop a rule to determine in advance which accident claims are likely to be fraudulent and which are likely to be non-fraudulent. Claims classified as potentially fraudulent can receive closer scrutiny than other

[1] And also contrasts with FA (factor analysis) for the same reason.

claims. If the insurer has a set of claims that it has investigated and found to be fraudulent and a set of non-fraudulent claims, the insurer can use the two groups as a training set to infer a rule for assigning new claims to the two groups. Essentially, the insurer would take the training set and "regress" whether or not the claims are fraudulent on a set of predictor variables and produce an equation to score new claims for fraudulent potential. New claims that receive high scores would be assigned to the fraudulent group; new claims that receive low scores to the non-fraudulent group. The insurer could then spend more effort checking out the putatively fraudulent claims. All this would be supervised learning.

By contrast, in CA, there would be no set of correctly assigned fraudulent and non-fraudulent claims. The insurer would just have a set of claims. The number of natural groups would be unknown (although ultimately the insurer wants to identify two groups: fraudulent and non-fraudulent). It would not even be known if any of the natural groups would correspond to fraudulent claims. Fraudulent claims could be dispersed randomly among whatever natural groups there may be. What a cluster analysis would do is to find clusters of similar claim characteristics so that the insurer could start to understand the different types of claims that it experiences. Some of those clusters may contain more fraudulent claims than other clusters. That may help with the specific problem of identifying likely fraudulent claims.

Example. An online content provider is often compensated by advertisers every time a viewer clicks on an ad displayed on a page seen by the viewer – whether or not the clicker subsequently buys anything. Clicking on an ad is called a "click-through", or CT. Content providers entice viewers to click through by bribing them with articles thought likely to be of interest ("click bait"). A bribe article is then surrounded or imbedded with related ads. For example, a viewer who clicks on an article about reducing car expenses may be presented an ad for a new car. The content provider has two predictions to make: Will the viewer click on the article? If the viewer clicks on the article, will the viewer click through to the ad?

Supervised learning can help by analyzing the training set of past data of articles and ads that viewers have or have not clicked on. There are two groups: CTs and non-CTs. The goal is to use the characteristics of clickers and non-clickers to predict the probability that future viewers will click through.

Unsupervised learning can help by discovering naturally occurring clusters among content viewers. E.g., finding a cluster of young males interested in sports and cars and another cluster of older females interested in fine clothes and beauty products suggests different articles and ads that might be presented to each cluster in order to increase the probability of a CT. If there is no cluster of viewers interested in movie stars, then that suggests de-emphasizing articles and ads that feature them.

Because of the lack of supervision, CA is a more difficult challenge than classification. There is not a great deal of theoretical foundation for the clustering procedures that have been proposed. Many clustering algorithms are *ad hoc* procedures that have been proposed because they merely *seem* intuitively plausible. There has been a fair amount of simulation and empirical work to try to learn how competing clustering methodologies are likely to work. These clustering methodologies have originated in many different disciplines, as the list of examples on page 1 of these notes suggests. CA became a hot topic after the publication of *Principles of Numerical Taxonomy* (1963) by biologists Robert Sokal and Peter Sneath. They introduced numerical algorithms for grouping species by common descent rather than by subjective judgment of anatomical similarities.

There are five basic steps in all CA studies:

1) Select the observations to be clustered.
2) Define the variables to use in clustering the observations.
3) Compute similarities (or dissimilarities) among the observations.
4) Create groups (clusters) of similar observations.
5) Validate the resulting clusters.

Cautions about CA:

1) Most clustering techniques are simple procedures without much underlying statistical theory.
2) Clustering techniques evolved from many disciplines and bear the biases of those disciplines.
3) Different clustering techniques yield different solutions for the same data set.
4) The strategy of CA is *structure-seeking*, but its operation is *structure-imposing*.

Terminology and Notation

*Definition: **Clusters** are continuous regions of variable space containing a relatively high density of data points separated from other high density regions by areas containing a relatively low density of data points.*

The things being classified are variously called **cases**, **observations**, **objects**, **entities**, **patterns**.

The aspects of the entities used to assess their similarities/dissimilarities are variously called **variables**, **attributes**, **characters**, **features**.

For example, in Figure 1, 19 companies are plotted by their profitability and growth rate. It is easy to see at least two clusters of companies. In the upper right are companies of high profitability and growth rate. In the lower left is a denser set of companies of low profitability and growth rate.

There are some characteristics of clusters that can be quantified when cases are plotted as points in *p*-dimensional variable space:

1) **Density** -- the concentration of cases is high within a cluster
2) **Variance** -- the dispersion from the center of a cluster is small for the cases in the cluster
3) **Dimension** -- the size or radius of the cluster is small
4) **Shape** -- clusters are typically ellipsoidal[2]
5) **Separation** -- the degree to which clusters overlap or are disconnected.

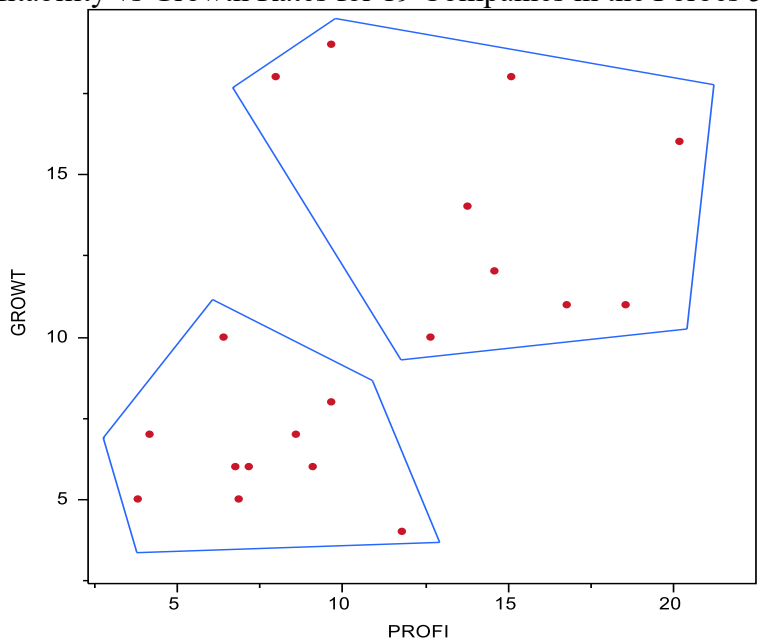
These characteristics can be used to help define algorithms to group objects into clusters.

We can think of the entities as the rows in a data matrix, with the attributes as the columns. Thus the data matrix corresponds to the standard R data table or Excel spreadsheet,

[2] Clusters that have the shape of a string or a ring are hard to identify.

with observations as rows and variables as columns. Let n denote the number of rows (entities) and p denote the number of columns (variables). The data matrix is n x p.

Figure 1. Profitability vs Growth Rates for 19 Companies in the Forbes 500



From the data matrix, a **similarity** (or **dissimilarity**) **matrix** will be computed. The entry in row i and column j of the similarity matrix shows how similar (dissimilar) entity i and entity j are. Thus, the similarity matrix must be n x n. Sometimes correlation is used to measure similarity. If correlation is used, the resulting similarity matrix is very different from the ordinary correlation matrix for the same n x p data matrix. The ordinary correlation matrix is p x p. The entry in row i and column j of the ordinary correlation matrix shows how similar *variable i* and *variable j* are. An example is the correlation between the rent and the area of a set of apartments. The set of all rents would be correlated with the set of all areas. By contrast, the entry in row i and column j of a similarity matrix shows how similar *entity i* and *entity j* are. An example would be the correlation between apartment 1 and apartment 2. The rent, area, age, number of rooms, etc. of apartment 1 would be correlated with the values of the same variables for apartment 2. This is an unusual and controversial type of correlation because the numbers being correlated are of very different units. An n x n similarity matrix shows how similar the rows of the n x p data matrix are, whereas the ordinary p x p correlation matrix shows how similar the columns of the n x p data matrix are. The similarity matrix is also called a **proximity matrix**. If desired, a dissimilarity matrix often can be created from a similarity matrix by simple means, such as subtracting correlations from ±1.

Similarities and Dissimilarities

Two concepts underlie most techniques for calculating (dis)similarities among cases that are to be clustered: (A) correlation and (B) distance.

(A) **CORRELATION COEFFICIENTS** for similarities. This is a strange kind of correlation coefficient. It is a correlation between observations instead of between variables. It uses the same formula as the ordinary correlation coefficient, but the formula is applied to two rows of the *n* x *p* data matrix instead of two columns. This results in mixing together numbers measured in possibly very different units to find means, standard deviations, and covariances.

> Ex: Jim's (age, salary, family size, home value) data vector of (40 years, $35000, 3 people, $80000) would be correlated with Jane's (age, salary, family size, home value) of (30 years, $50000, 4 people, $60000) to yield 0.9295166 as the similarity between Jim and Jane.

Besides the issue of potentially meaningless units resulting from composites across unrelated variables, correlations as similarities have the difficulty that small variations in large numbers (such as salary) may overwhelm proportionately much larger variations in smaller numbers (such as family size) in the similarity measure; therefore, the clustering may be driven by, and cater to, variables with large values, such as salary.

> Ex: In the preceding example, if Jane's family size increases by 25%, her correlation with Jim increases microscopically to 0.9295170. But if her salary increases by 25%, her correlation increases to 0.9736926.

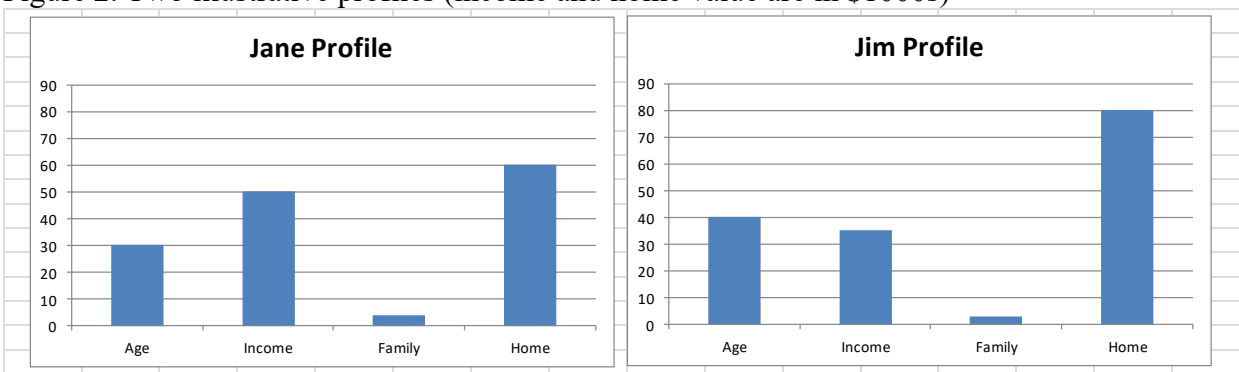
An obvious possible remedy is to standardize all variables before clustering, but this suggestion is not universally endorsed because it implies that all variables are equally important in assessing similarity. Another potential problem results from the selection of highly correlated variables as the basis for clustering. For example, if X_1, X_2, X_3 are three highly correlated variables used with an unrelated X_4 to compute a similarity measure, the effect is almost the same as using just X_1 and X_4, but giving X_1 three times the weight of X_4. So selecting all four variables to use for CA could bias results. Principal Components Analysis can be used prior to CA to reduce the number of variables and produce uncorrelated "variables", but there is controversy over this, as well, because PCA may tend to blur together two distinct clusters that are actually widely separated.[3] The problems discussed in this paragraph are also faced by distance (dissimilarity) measures discussed in (B), below.

Profile is a useful concept for comparing entities. A profile of a case is simply a vector of attribute values for a case -- in other words, the profile of a case is its row in the *n* x *p* data matrix for that case. A profile is usually displayed as a graph with the variable names listed on the horizontal axis and the scores (values) of the variables shown vertically as bars and/or connected dots. CA can be thought of as a way to group cases with similar profiles.

[3] Using the complete set of principal components preserves the geometry and therefore the clusters. However, using a subset of principal components changes the geometry.

Ex: Figure 2 visually shows the profiles for Jane and Jim. The variables are forced to use the same vertical scale although their units are quite different.

Figure 2. Two illustrative profiles (income and home value are in $1000s)



The similarity between profiles can be decomposed into three parts:

1) **Shape** -- the pattern of dips and rises of the profile as you look across the variables
2) **Scatter** -- the dispersion of the scores around their average
3) **Elevation** -- the mean score of the case over all of the variables.

The correlation coefficient can be enlisted to measure the similarity between two profiles. But it is sensitive only to *shape*. It is unaffected by differences between two profiles in terms of *scatter* and/or *elevation*. Two profiles with the same shape but different scatter and/or elevation will have correlation coefficient = 1.00. If you add the same number to every component of a profile, and/or if you multiply every component of a profile by the same positive number, the correlation with any other profile remains unaffected.

Correlation measures for similarities are seldom used in practice.

(B) **DISTANCE MEASURES** for dissimilarities. These are the most common (dis)similarity measures. Common measures of distance between cases are:

1) Euclidean distance (L^2 distance) = $\sqrt{\sum_{k=1}^{p}(x_{ik}-x_{jk})^2}$
2) "Manhattan" (city-block – L^1) distance = $\sum_{k=1}^{p}|x_{ik}-x_{jk}|$
3) Mahalanobis distance = $(\mathbf{x}_i-\mathbf{x}_j)'\mathbf{S}^{-1}(\mathbf{x}_i-\mathbf{x}_j)$

where x_{ik} is the score of case *i* on variable *k*, x_{jk} is the score of case *j* on variable *k*, $(\mathbf{x}_i-\mathbf{x}_j)'$ is the 1 x *p* row vector of differences between row *i* and row *j* of the data matrix, and $\mathbf{S}^{-1}$ is the inverse of the *p* x *p* covariance matrix of the *variables* (not the cases).[4]

[4] Note that $(\mathbf{x}_i-\mathbf{x}_j)'\mathbf{S}^{-1}(\mathbf{x}_i-\mathbf{x}_j)$ is conformable. It is (1 x p) x (p x p) x (p x 1) = 1 x 1.

Ex: Given Jim's (age, salary, family size, home value) data vector of (40 years, $35000, 3 people, $80000) and Jane's (age, salary, family size, home value) of (30 years, $50000, 4 people, $60000):

1) The Euclidean distance between Jim and Jane = sqrt((40-30)^2 + (35000-50000)^2 + (3-4)^2 + (80000-60000)^2) = 25000;
2) The "Manhattan" distance = abs(40-30) + abs(35000-50000) + abs(3-4) + abs(80000-60000) = 35011;
3) Calculation of the Mahalanobis distance between Jim and Jane needs data on the other cases in order to populate the cells in $\mathbf{S}^{-1}$.

Generally, when Euclidean distance is used as a (dis)similarity measure, the square root is not taken because of the relative ease of analyzing sums of squares and the difficulty of analyzing square roots. Mahalanobis distance is a generalization of Euclidean distance that adjusts for different variances and correlations between the variables. Euclidean distance and Mahalanobis distance are identical if the variables are uncorrelated and have the same variances. [5]

In addition to the problems in common to correlation and distance measures mentioned in (A) above, distance measures are strongly affected by the scatter (scale) of variables used in profiles. Therefore, it is standard practice to standardize variables prior to calculating distance measures used as (dis)similarities.

R has separate procedures for different clustering variants. For example, `hclust` for hierarchical tree-growing, and `kmeans` for iterative partitioning.

Validating the Cluster Analysis

Like the determination of the number of clusters, validating the grouping of cases into clusters is an unsolved problem. Two methods in common use are rejected by theoreticians and/or fail simulation studies:

(1) **Cophenetic correlation** (only for hierarchical agglomerative methods). This is the ordinary correlation between the entries in the original similarity matrix and an artificial similarity matrix constructed from the tree: The latter matrix uses the height (similarity) of the tree where two entities have their earliest common ancestor as the constructed similarity between the two entities. Thus cophenetic correlation measures how well the tree actually represents the similarities between entities. But there are only n - 1 possible distinct values for the constructed similarity matrix, whereas the original similarity matrix has n (n - 1)/2. And it assumes normality. And at least one simulation found it wanting.

(2) **(M)ANOVA F-statistics** are commonly used but theoretically invalid for the reasons cited above under "Determining the Number of Clusters".

Three potentially useful validation techniques:

(3) **Cross-validation**. Split the cases and run CA on each set. If similar clusters develop, then the CA may be OK; but if different clusters appear, then the CA is not generalizable. (Note that this method is a negative method in that it can disconfirm but not confirm the CA.) A variation on this method is to run the CA on the first set of cases, then use its cluster centroids as

[5] In that case, $\mathbf{S} = \mathbf{I}$. If Euclidean distance may be thought of as distance between nested spheres, Mahalanobis distance may be thought of as distance between nested ellipsoids.

seeds to cluster the second set. This forces the same number of clusters in the cross-validation. If the cluster centroids from the first set reproduce similar case assignments and the clusters in the second set of cases have small within variation and big between variation, then the CA looks good.

(4) If the clusters are separating the cases on the variables used in the CA, then they should separate the cases on other variables NOT used to develop the CA. Try it with new variables. Since such variables could have been used to develop the CA, this suggestion is somewhat like cross-validation by splitting the variables instead of the cases.

(5) **Simulation**. Create a simulated data set with major characteristics like those of the real data. Run CA on each and compare results.

Interpretation of Clusters

Like the interpretation of principal components (PCA) or factors (FA), the interpretation of clusters is an art. Nevertheless, some valuable guidance can be given:

First, since the objective of CA is to group cases into entities called clusters that are similar within clusters but different between clusters, one can treat a cluster as a single entity for interpretation. This is usually done by choosing the centroid to represent each cluster. That is, all of the cases within a cluster are assigned the same interpretation, provided by the centroid.

Second, one usually looks to the vector of means of the clustering variables and the vector of counts of the number of cases within each cluster to define meaning for each cluster.

Third, one builds a "story" about the cluster based upon the vector of cluster variable means. One imagines an exemplar case that represents the cluster. The exemplar case is imagined to have the vector of clustering means as its actual values. The story is about the exemplar case: On which variables is it high and on which is it low? The story takes the other clusters into account by emphasizing how the exemplar case differs from the exemplar cases of other clusters. Emphasize the contrast between exemplars.

Fourth, in building the stories, it may be important to add the relative size (number of cases) of each cluster to the story being built. This may not be important if the clusters are of roughly the same size. But if they differ substantially in size, it is usually important to know that.

Fifth, in order to make the cluster stories memorable, one searches for a one-word or short-phrase characterization of each story. This is not always possible. But when it is possible in a business setting, it can give management a valuable tool. For example, a cluster analysis of the customers of a financial services firm might find that they consist of two groups: a large group of retired executives and a smaller group of well-paid corporate officers. The firm may have been marketing its services at corporate employees. But over time, it may be that those employees have been retiring and keeping their association with the firm. This suggests the possibility of expanding the scope of the financial services offered by the firm to include specific products and services aimed at retired executives. Having a memorable story about the firm's customers is clarifying for explaining simply the firm's mission to its employees and to the public.